Patrick Léonnel Nzudom

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Education

Polytechnique Montréal Jan. 2023 – May 2027 (expected)
Bachelor of Engineering (B.Eng.), Computer Engineering · Montreal, QC

Universidad de Sevilla – Escuela Técnica Superior de Ingeniería Informática Jan. 2026 – June 2026
Exchange semester, Computer Engineering · Seville, Spain

Research Interests

Embedded and cyber-physical systems, distributed and edge computing, network and systems security.

Selected Coursework

Grade A: Computer Systems Design Project (INF3995), Distributed Systems and Cloud Computing (INF8480), Fixed and Mobile Network Security (INF8402), Computer Security (INF4420A), Files and Databases (INF3710), Web Application Software Project (LOG2990, A*)
Other relevant: Embedded Systems (INF3610), Embedded Systems: Design and Verification (INF8500), Parallel Computing Systems (INF8601), Digital Systems Design (INF3500), Computer Networks (INF3405), Operating Systems (INF2610)
In progress: Computer Engineering Capstone Project (INF8970), Data Mining (INF8111)

Experience

Web Developer Intern — Voyages en Direct Oct. 2026 – Present
Montreal, QC

- Deploy and maintain web services on Kubernetes using a GitOps workflow and CI/CD pipelines.
- Work on site performance, SEO and baseline web security.

Embedded Software Developer Intern — Neptronic Sept. 2025 – Dec. 2025
Montreal, QC

- Developed automated tests for embedded touchscreen software using Robot Framework.
- Worked with the BACnet and Modbus building-automation communication protocols.

Embedded Software Developer Intern — Neptronic May 2025 – Aug. 2025
Montreal, QC

- Contributed to the design, development and documentation of embedded engineering software.
- Refactored and optimized existing applications to improve performance and efficiency.
- Developed and validated regression tests directly on embedded hardware.

Projects

Multi-Robot Exploration System Sept. 2025 – Dec. 2025
ROS2, AgileX Limo, Docker

- Team design project (INF3995) building an autonomous exploration system on AgileX Limo robots with ROS2.
- Followed a formal engineering review cycle: Preliminary Design Review, Critical Design Review and Readiness Review.

Babillard – Student Community Platform Oct. 2026 – Present
Next.js, Supabase, Vercel · notrebabillard.ca

- Designed and built a web platform centralizing events and announcements from Montreal university student associations.
- Added a sublet and temporary roommate module aligned with university terms and Quebec rental rules (TAL).
- Implemented authentication and a secured database with Supabase, with continuous deployment on Vercel.

Embedded Systems Project Jan. 2024 – Apr. 2024
C++, AVR ATmega324PA, AVR GCC · link

- Programmed two robots communicating over infrared and RS232 links on an ATmega324PA microcontroller.
- Integrated peripherals: piezo buzzer, line-following sensor and infrared sensor.

Tactical Role-Playing Game Platform Jan. 2025 – Apr. 2025
TypeScript, Angular, NestJS · Ram Raider

- Built a multiplayer web game in a team using Agile practices, with an emphasis on testing and quality assurance (final grade A*).

Exploring Markov Chains Jan. 2022 – Apr. 2022
Python, LaTeX · link

- Simulated Markov chain applications in Python and reported experimental results and analysis in LaTeX.

Leadership and Outreach

Web Developer — PolyAI Technical Society — Polytechnique Montréal, polyai.ca June 2024 – Present
Montreal, QC

- Develop the society's website (TypeScript, Angular, NestJS) and organize learning events such as the Learnathon.

Science Facilitator — École Pie-XII Oct. 2021 – May 2022
Montreal, QC

- Introduced children aged 5 to 12 to programming with ScratchJr.

Technical Skills

Languages: C/C++, Python, Java, TypeScript/JavaScript, SQL, VHDL, Assembly
Embedded & Protocols: AVR microcontrollers, ROS2, BACnet, Modbus, RS232
Systems & DevOps: Linux, Docker, Kubernetes, GitOps, CI/CD, Git
Web & Data: React, Next.js, Angular, Node.js, NestJS, Supabase, MySQL, SQL Server, MongoDB, pandas, NumPy
Testing: Robot Framework, JUnit, Jest (TDD)
Languages spoken: French, English